

PROFESSIONAL PROGRAM IN

DATA ANALYTICS

with Agentic AI Immersion

Master SQL, Python, statistics and BI — then build with generative AI and design agentic workflows that run analysis end to end.



6 Months
Live + Applied



26 Modules
One Clear Path



10+ Projects
Real-World Work



5 Weeks
Agentic AI

Professional Program in Data Analytics

CareerZeta's Data Analytics Professional Program is a six-month, live and applied program that takes you from spreadsheets to SQL, Python, statistics, BI and modern data engineering — then adds a five-week Agentic AI Immersion so you can build AI agents that run analysis end to end.

 Certificates from NASSCOM FutureSkills Prime and Microsoft

REVENUE +18.4%



segment	revenue
enterprise	4.82M
mid-market	2.37M
smb	1.19M
consumer	0.86M

Why Data Analytics?



Every Industry Runs on Data

Banking, healthcare, retail, telecom, logistics and IT all hire analysts to turn raw data into decisions.



Skills That Travel

SQL, Python, statistics and BI work in every sector, so your options stay open as you grow.



From Database to Dashboard

Analysts who can work end to end, without waiting on an engineering team, are the ones companies compete for.



A Career Ladder, Not Just a Job

Grow from Data Analyst to Senior Analyst, Analytics Engineer, Analytics Lead and Head of Data.



AI Makes Analysts Faster

Agentic workflows now automate the repetitive parts of analysis. Analysts who can build them move ahead.



Proof You Can Show

Graduate with real projects and a portfolio an employer can open — not just a certificate.

Program at a Glance



6 Months

Live + applied learning



26 Modules

19 core + 7 Agentic AI



8 Phases

Foundations to applied analytics



10+ Projects

Real-world, portfolio-ready



6 Case Studies

Across six industries



5 Weeks

Agentic AI Immersion

About Program

The Data Analytics Professional Program is built for people who want to work directly with data — query it, clean it, model it and present it — and to use AI agents to do that faster. Across eight phases you move from spreadsheet analytics to SQL, Python, statistics, BI and the modern data stack, finishing with applied analytics, a capstone and a five-week Agentic AI Immersion.



Learning Format
Live + Applied



6 Months
Live Program



Career Services
By CareerZeta



Certification
NASSCOM & Microsoft

Key Highlights

- ✓ 26 modules across 8 phases
- ✓ 10+ real-world projects
- ✓ 6 industry case studies
- ✓ 5-week Agentic AI Immersion
- ✓ Certificates from NASSCOM FutureSkills Prime & Microsoft
- ✓ Designed for working professionals & graduates
- ✓ Live sessions with industry practitioners
- ✓ 1:1 mentoring sessions
- ✓ Resume & LinkedIn profile building
- ✓ Mock interview preparation
- ✓ Hackathons & peer network
- ✓ No-cost EMI options

Program Pedagogy



Live Instructor-led Sessions

Learn from practitioners who work with data every day.



Hands-on Projects

Build 10+ projects on real-world datasets.



Industry Case Studies

Work through six cases across finance, telecom, healthcare and more.



1:1 Mentoring

Regular check-ins with a mentor to guide your transition.



Hackathons

Solve real problems in teams and showcase your skills.



Peer & Alumni Network

Learn with a cohort and stay connected for referrals and support.



Portfolio Review

Mentor feedback on your capstone and projects before you apply.



Agentic AI Labs

Build, test and govern AI agents that automate analysis.

Who Is This **Program For?**

- ✓ Working professionals moving into a data analyst role
- ✓ Excel-heavy analysts ready to step up to SQL, Python and BI
- ✓ IT, operations and business professionals who work with data every day
- ✓ Graduates who want a hands-on start in analytics
- ✓ Anyone who wants to build AI agents into their analysis work

How to **Join**

Four steps take you from application to your first live session.

1

SUBMIT YOUR APPLICATION

Tell us about your background and why you want to move into data analytics.



2

SCREENING CALL

A short conversation with our admissions team to understand your goals and confirm program fit.



3

ENROLLMENT & ONBOARDING

Secure your seat, get access to pre-work, and meet your cohort before day one.



4

KICK OFF MODULE 01

Join your first live session and start building from week one.



Flexible Payment

No-cost EMI options available



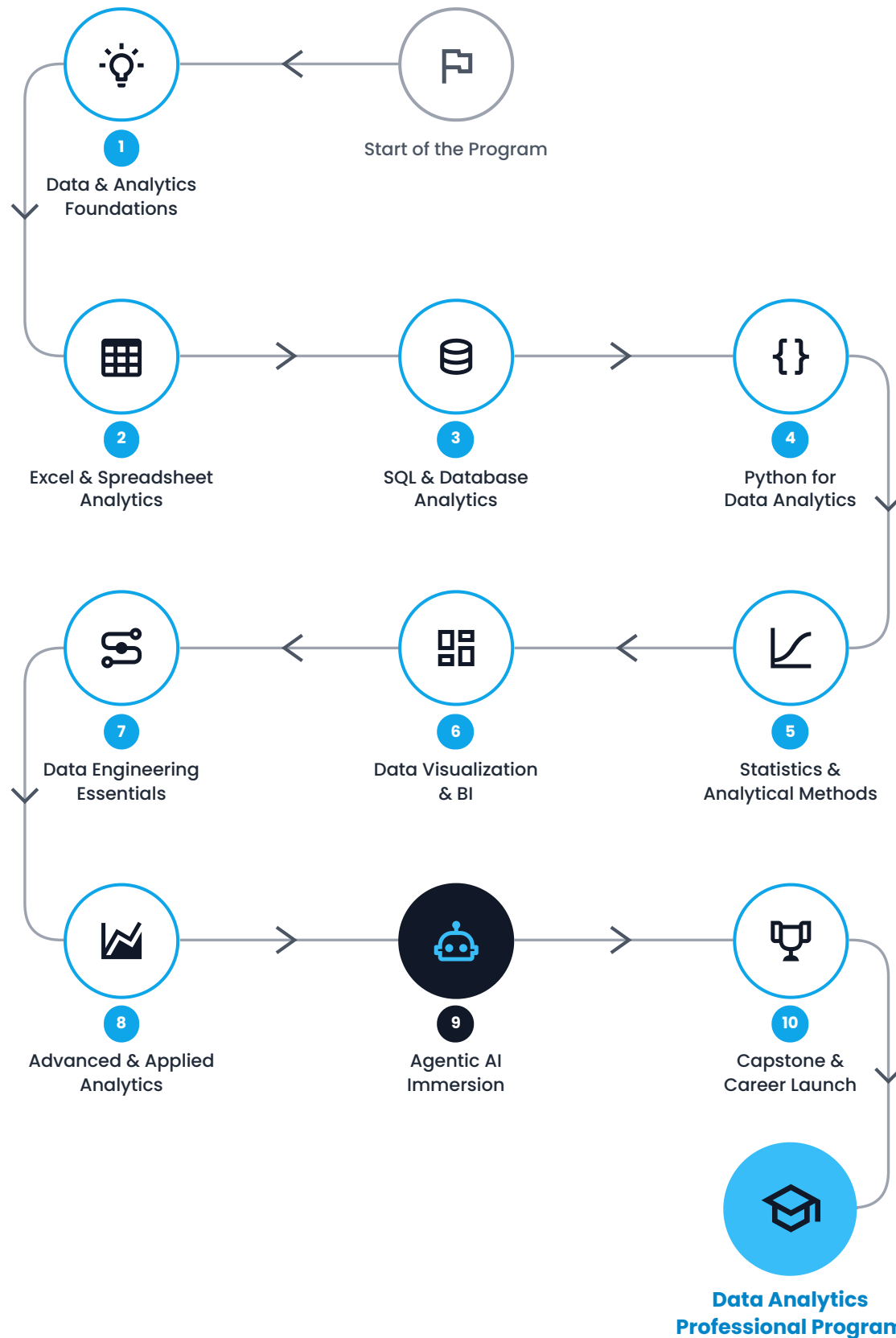
Cohort Size

Kept small for real mentor access

Your Journey

○ Core phases

● Agentic AI



PHASE 1 · DATA & ANALYTICS FOUNDATIONS

Module 01

Introduction to Data Analytics

- What is Data Analytics?
- Data Analytics vs Business Analytics vs Data Science
- Types of Analytics: Descriptive, Diagnostic, Predictive & Prescriptive
- The Analytics Lifecycle: Question → Data → Analysis → Insight
- Data-Driven Decision Making
- Metrics, KPIs & Measurement
- Role of a Data Analyst
- Analytics Team Workflows

Module 02

Data Literacy, Data Types & Data Quality

- Types of Data
- Structured, Semi-Structured & Unstructured Data
- Data Sources & Collection Methods
- Data Quality Dimensions
- Data Profiling & Validation
- Handling Missing & Inconsistent Data
- Data Documentation & Dictionaries
- Data Privacy, Ethics & Responsible Use

PHASE 2 · EXCEL & SPREADSHEET ANALYTICS

Module 03

Excel for Data Analysis

- Excel Fundamentals
- Data Cleaning & Transformation
- Formulas & Functions
- Lookup Functions
- Conditional Functions
- Pivot Tables & Pivot Charts
- Data Validation
- Charts & Dashboards
- Statistical Functions in Excel

Module 04

Power Query & Advanced Excel Automation

- Power Query Interface
- Importing from Multiple Sources
- Merge & Append Queries
- Data Shaping & Unpivoting
- Power Pivot & Data Models
- Introduction to DAX in Excel
- Automated Refresh & Reporting
- Building Repeatable Data Workflows

PHASE 3 · SQL & DATABASE ANALYTICS

Module 05

SQL Fundamentals

- Databases & Relational Data
- SELECT, WHERE, ORDER BY
- Filtering & Sorting
- Aggregations
- GROUP BY & HAVING
- Joins
- Subqueries
- CASE Statements
- Data Types & Functions

Module 06

Advanced SQL & Query Optimization

- CTEs
- Window Functions
- Ranking & Segmentation
- Date & Time Analysis
- Advanced Joins & Set Operations
- Cohort & Funnel Analysis
- Views & Stored Procedures
- Indexing & Query Performance
- Writing Production-Grade Queries

PHASE 4 · PYTHON FOR DATA ANALYTICS

Module 07

Python Programming Foundations

- Python Setup & Notebooks
- Variables, Data Types & Operators
- Lists, Dictionaries, Tuples & Sets
- Conditionals & Loops
- Functions & Modules
- Error Handling
- File Handling
- Writing Clean, Reusable Code

Module 08

Data Wrangling & Exploratory Analysis

- NumPy Arrays & Vectorization
- Pandas Series & DataFrames
- Indexing, Filtering & Slicing
- Handling Missing Data & Duplicates
- Merging, Joining & Reshaping
- GroupBy & Aggregation
- Exploratory Data Analysis Workflow
- Matplotlib & Seaborn
- End-to-End Python EDA Project

PHASE 5 · STATISTICS & ANALYTICAL METHODS

Module 09

Descriptive Statistics & Probability

- Mean, Median & Mode
- Variance & Standard Deviation
- Probability Fundamentals
- Distributions
- Normal Distribution & Z-Scores
- Sampling & Sampling Methods
- Correlation & Covariance
- Outlier Detection

Module 10

Inferential Statistics & Hypothesis Testing

- Population vs Sample
- Central Limit Theorem
- Confidence Intervals
- Hypothesis Testing & p-values
- t-test, Chi-square Test & ANOVA
- Regression Fundamentals
- A/B Test Design & Analysis
- Statistical vs Practical Significance
- Interpreting Results for Stakeholders

PHASE 6 · DATA VISUALIZATION & BI

Module 11

Power BI & Data Storytelling

- Power BI Ecosystem
- Data Import & Power Query
- DAX Fundamentals
- Calculated Columns & Measures
- Time Intelligence
- Interactive Reports & Dashboards
- Choosing the Right Chart
- Storytelling with Data
- Power BI Service & Sharing

Module 12

Tableau

- Tableau Fundamentals
- Connecting & Blending Data
- Dimensions & Measures
- Calculated Fields & LOD Expressions
- Filters & Parameters
- Maps & Geographic Analysis
- Dashboards & Actions
- Interactive Visualizations
- Avoiding Misleading Visualizations

PHASE 7 · DATA ENGINEERING ESSENTIALS

Module 13

Data Warehousing, ETL & Pipelines

- Data Warehouse Concepts
- OLTP vs OLAP
- Star & Snowflake Schemas
- Fact & Dimension Tables
- ETL vs ELT
- Data Pipeline Fundamentals
- Workflow Orchestration Concepts
- Data Quality in Pipelines

Module 14

Cloud Analytics & the Modern Data Stack

- Cloud Data Platforms Overview
- Cloud Storage & Compute Basics
- Cloud Data Warehouses
- Connecting BI Tools to Cloud
- Version Control with Git & GitHub
- Collaboration & Analytics Workflows
- Access, Cost & Governance Basics
- The Modern Data Stack End to End

PHASE 8 · ADVANCED & APPLIED ANALYTICS

Module 15

Predictive Analytics & ML Foundations

- Introduction to Predictive Analytics
- Supervised vs Unsupervised ML
- Linear & Logistic Regression
- Decision Trees & Random Forest
- Clustering & Segmentation
- Feature Engineering & Selection
- Model Evaluation Metrics
- Model Interpretation

Module 16

Time Series & Forecasting

- Time Series Components
- Trend & Seasonality Decomposition
- Moving Averages & Smoothing
- Stationarity & Differencing
- ARIMA & Exponential Smoothing
- Forecast Accuracy Metrics
- Demand & Revenue Forecasting
- Communicating Uncertainty

Module 17

Customer & Marketing Analytics

- Customer Segmentation
- Cohort & Retention Analysis
- Customer Lifetime Value
- Campaign & Channel Analysis
- Marketing Funnel & Conversion
- Customer Acquisition Cost
- Product & Behavioural Analytics
- Web & Digital Analytics

Module 18

Financial & Operational Analytics

- Financial KPIs & Reporting
- Profitability & Cost Analysis
- Budget vs Actual Analysis
- Revenue & Pricing Analytics
- Inventory & Supply Chain Analytics
- Operational KPIs
- Risk & Fraud Analytics
- Process & Efficiency Analysis

Module 19

Capstone & Analytics Portfolio

- End-to-End Capstone Project
- Problem Framing & Data Sourcing
- Analysis, Modelling & Validation
- Dashboard & Report Delivery
- Presenting to Stakeholders
- Building an Analytics Portfolio
- Interview-Ready Walkthroughs



5-Week Agentic AI Immersion

Seven modules on designing, connecting and governing AI agents — pointed at the work analysts actually do: querying, cleaning, analysing and reporting.

Module 1

Agentic AI Fundamentals

- Basics of Machine Learning and Generative AI
- Agent architecture
- Agents vs chatbots vs automation
- Single vs multi-agent design
- When to use/not use agents

Module 2

AI Use-Cases Across the Analytics Workflow

- Mapping the analytics workflow
- AI opportunity identification
- Task decomposition
- Automation assessment
- Human vs AI responsibilities
- Agent feasibility
- ROI and the analytics business case

Module 3

Prompt & Context Engineering

- Prompt design
- Structured prompts
- Few-shot prompting
- Prompt templates for SQL and Python
- Context engineering
- Schemas as context

Module 4

Tools, APIs & MCPs

- Tools orchestration
- APIs
- MCPs
- Connecting agents to warehouses and BI tools
- Database/API integration

Module 5

RAG for Analytics

- RAG architecture and pipeline
- Filtering types
- Vector databases and Embeddings
- RAG over data dictionaries
- RAG evaluation

Module 6

Agent Memory, Planning & Workflows

- Memory management
- Planning
- Task decomposition
- Sequential workflows : Chain of thought
- Conditional workflows
- Agent loops
- Human in the loop
- Multi-agent orchestration

Module 7

Agent Evaluation, Guardrails & Governance

- Agent evaluation
- Hallucination checks on results
- Guardrails
- Observability
- Data governance and access control

Skills to Master

- SQL & Query Optimization
- Excel & Power Query
- Python, pandas & NumPy
- Exploratory Data Analysis
- Statistics & A/B Testing
- Power BI & DAX
- Tableau
- Data Storytelling
- Data Warehousing & ETL
- Cloud Analytics
- Forecasting & Predictive Models
- Agentic AI Workflows

AI & Data Tools

Gain hands-on experience with industry-leading tools and technologies that power Data Science, AI and Analytics across the globe.



...and many more!

Projects and case studies cover the following industries:



Finance



Healthcare



Retail



E-commerce



Marketing



Telecom



Logistics



Pharma

Beginner

Complex SQL Sub-Queries

Solve layered database problems using SQL subqueries. Work with IN, NOT IN, EXISTS, NOT EXISTS, ANY and ALL to filter, compare and retrieve data efficiently. This project builds the query fluency that almost every data analyst interview screens for.

Beginner

Query a Large Relational Database

Work with large relational datasets and learn to retrieve meaningful customer information using SQL. Apply tables, data types, functions and operators to perform effective queries at scale. This project strengthens foundational database skills while showing how businesses organise, access and analyse large volumes of structured data.

Intermediate

Data Cleaning & Preparation with Python

Take a messy real-world dataset and make it analysis-ready with Pandas. Handle missing values, duplicates, inconsistent formats and outliers, then document a repeatable cleaning workflow other analysts can rerun on next month's data.

Intermediate

Exploratory Data Analysis on Real-World Data

Run a full exploratory analysis on an unfamiliar dataset. Profile distributions, test correlations, surface anomalies and build a findings notebook with Matplotlib and Seaborn. The project develops the habit of interrogating data before modelling it, and of turning what you find into questions a business can act on.

Intermediate

Customer Segmentation & Personalization

Segment customers using demographic, geographic and behavioural data. Build distinct groups, validate them statistically, and translate the segments into targeting, offer and personalisation recommendations a marketing team can act on.

Advanced

E-commerce Funnel & Cohort Analysis

Analyse user journeys from first visit to repeat purchase. Build cohort retention tables and funnel drop-off analysis in SQL, then quantify where and why customers leave. This project develops practical skills in retention, conversion and revenue analytics, and in presenting findings that product and growth teams can act on.

Advanced

Sales Forecasting & Demand Planning

Build time series forecasts on historical sales data. Decompose trend and seasonality, compare forecasting approaches, evaluate accuracy, and communicate uncertainty to planning stakeholders who have to commit budget to the numbers. This project covers demand planning, inventory signals and revenue projection end to end.

Advanced

Risk Management & Fraud Detection

Build analytical models to evaluate credit, fraud and operational risk. Identify unusual patterns and suspicious activity using anomaly detection techniques. This project introduces learners to risk assessment and fraud analytics while demonstrating how organisations use data to spot threats early and support faster, safer decisions.

Advanced

End-to-End ETL Pipeline & Data Warehouse

Design a star-schema warehouse and build the pipeline that feeds it. Extract from multiple sources, transform and load on a schedule, then connect a BI layer on top. This project puts the modern data stack together in miniature and shows how analysts work with the systems that supply their data rather than waiting on them.

Advanced

Executive KPI Dashboard

Create interactive dashboards in Power BI and Tableau to monitor revenue, margin, retention and operational health. Track trends, flag anomalies, and present insight in a form that shortens management decision cycles. This project develops practical skills in KPI monitoring, data visualisation and executive reporting.

Capstone & **Portfolio**

The program closes with an end-to-end capstone: frame a business question, source and model the data, ship a dashboard and report, present it to stakeholders — then shape it all into a portfolio an employer can open.

- ✓ End-to-End Capstone Project
- ✓ Problem Framing & Data Sourcing
- ✓ Analysis, Modelling & Validation
- ✓ Dashboard & Report Delivery
- ✓ Presenting to Stakeholders
- ✓ Building an Analytics Portfolio
- ✓ Interview-Ready Walkthroughs

```
df_csv("sales_2026.csv", parse_dates=["date"])
df = df_csv(subset=["region"]).drop_duplicates()
df.groupby(df.date.dt.to_period("M"))["revenue"]
risk"] = model.predict_proba(X_test)[: , 1]
plt.subplots(figsize=(12, 5))
```

```
SELECT c.segment, DATE_TRUNC('month', o.order_date) AS month,
SUM(o.order_value) AS revenue, COUNT(o.order_id) AS orders
FROM orders o JOIN customers c ON c.customer_id = o.customer_id
WHERE o.order_date >= DATE '2026-01-01'
GROUP BY c.segment, month
```



Six case studies put the full toolkit to work on the kinds of problems data analysts are hired to solve — each one framed around a real industry question and the decision it has to support.



FINANCE

Credit Risk Analysis

Analyse customer financial data to assess creditworthiness and repayment risk. Build and validate risk models that help lenders make faster decisions, reduce defaults and strengthen portfolio performance through data-driven credit evaluation.



MARKETING

Marketing Campaign Optimization

Evaluate campaign data across digital channels to understand what drives engagement and conversion. Attribute performance across audiences and channels, then reallocate budget to lift return on spend.



TELECOM

Customer Churn & Retention

Analyse usage, billing and service data to identify customers at risk of leaving. Build churn indicators, quantify the revenue at risk, and design retention interventions that hold up when they reach real customers.



HEALTHCARE

Patient Flow Optimization

Analyse patient arrival, treatment and discharge data to locate workflow bottlenecks. Optimise staffing, scheduling and resource allocation to reduce waiting times and improve the patient experience.



E-COMMERCE

Product & User Behaviour Analytics

Analyse clickstream and transaction data to understand how users move through a product. Measure feature adoption, diagnose where journeys break down, and prioritise the changes worth building next.



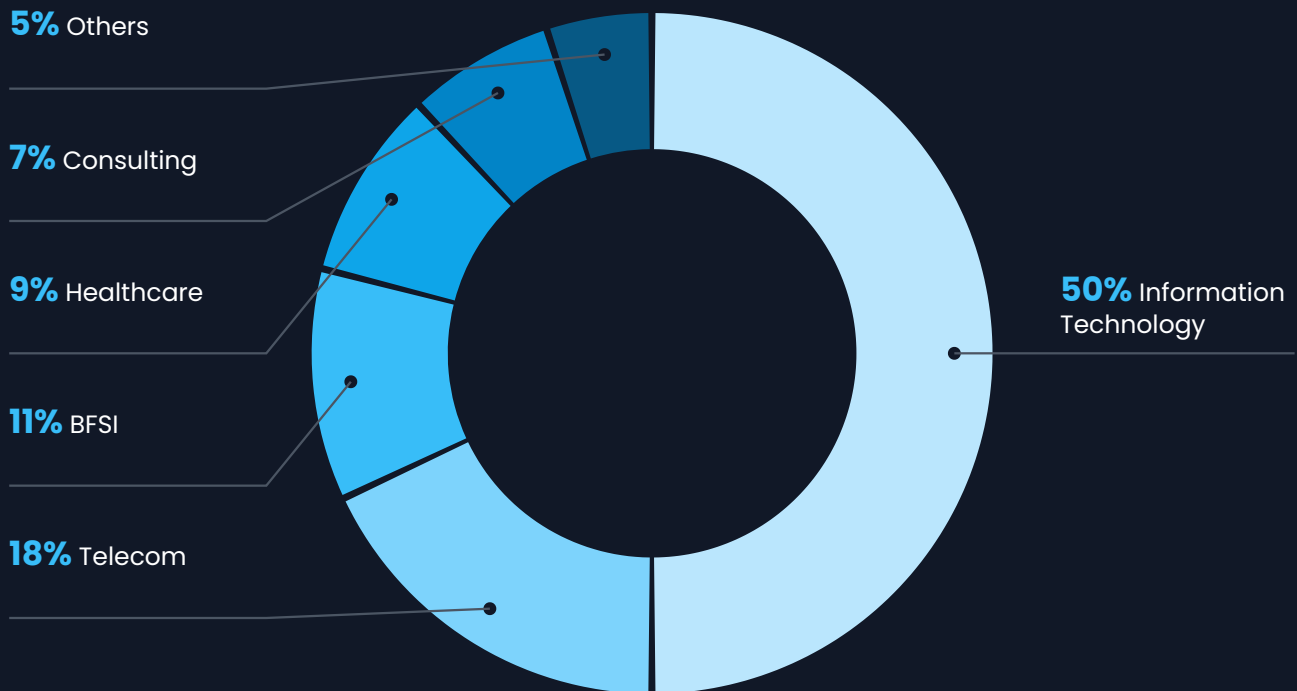
PHARMA

Pharma Sales & Demand Forecasting

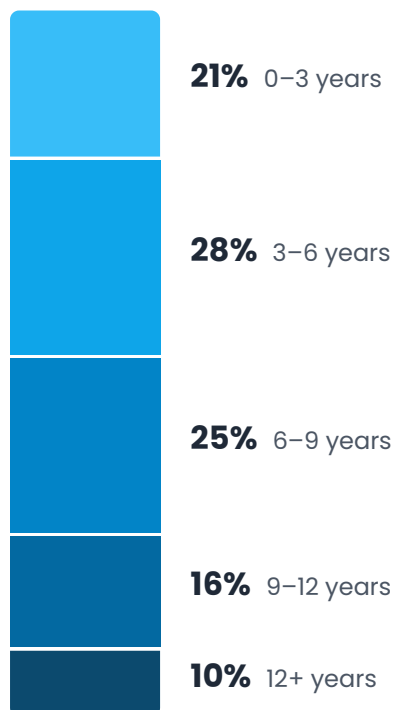
Analyse historical sales, prescription and market data to forecast product demand. Identify regional trends and shifting patterns to support inventory planning and distribution.

Meet the **Batch**

Industries Our Learners Come From



Work **Experience**



Some of their current employers include

amazon

citi

Capgemini

BOSCH

pwc

EY

accenture

Adobe

TATA
TATA CONSULTANCY SERVICES

vmware

Abbott

DELL

CareerZeta Career Services



1:1 Mentoring



Mock Interviews



Portfolio Review



Placements

What We Offer



1:1 Mentoring Sessions

Regular check-ins with a mentor to guide your transition into data analytics.



Resume & LinkedIn Profile Building

Craft a resume and profile built around the projects you actually shipped in this program.



Hackathons

Work on real-world problems, build solutions in teams, and showcase your skills to industry mentors.



Mock Interview Preparation

Practice with the questions employers actually ask data analytics candidates.



Career-Oriented Sessions

Live sessions on negotiation, career trajectory planning, and workplace communication.



Portfolio Review

Mentor feedback on your capstone and project portfolio before you start applying.



Alumni & Peer Network

Stay connected to a growing community of CareerZeta learners for referrals and support.



Placement Support

Get access to hiring partners, job opportunities and dedicated support to help you land the right role.

Top Destinations

```
hiring = ["Amazon", "Uber", "Microsoft", "KPMG"]
you.apply(hiring, portfolio=True, mock_prep=True)
support = CareerZeta.placement_support()
# resume + portfolio + practice = interview-ready
print("Next stop: your new ID badge.")
```

Companies across technology, banking, consulting, consumer and fintech that hire for the skills this program builds.



Top Faculty with CareerZeta

```
from careerzeta.faculty import Practitioners
mentors = Practitioners(day_jobs=True)
you.learn_from(mentors, live=True, doubts="all")
# war stories > textbook theory, every time
print("Mentor review: approved. Ship it!")
```

Learn from practitioners who build with data, engineering and AI in their day jobs.



Arnab Das
Data Engineer
Ieng Group



Anas Qureshi
Technical Lead
NetSkope



Devesh Thapliyal
Technology Lead
Infosys



Ashish Dahiya
Sr. Consultant
Centric Consulting



Rohit Bhatt
Data Management
Engineer
Zurich Insurance



Dileep KH
ML Engineer
University of the West
of England



Rajendra Dhama
Technical Lead
HCL Tech



Divyesh Pandey
Technical Lead
Accenture



Ayush Sharma
Sr. Manager
Angel One

About **CareerZeta**

CareerZeta is a global edtech platform committed to empowering professionals with industry-relevant, future-ready skills. We deliver practical, career-focused learning experiences designed to help working professionals upskill, advance their careers, and confidently transition into new roles and domains.

Our programs combine expert-led learning, hands-on projects, industry insights, and cutting-edge technology to help professionals stay ahead in today's rapidly evolving digital economy.

Whether you are looking to strengthen your expertise, take the next step in your career, or switch to a new professional profile, CareerZeta provides the skills, guidance, and practical exposure needed to move forward with confidence. We believe professional learning should be practical, accessible, and outcome-focused — helping individuals turn their ambitions into relevant skills, greater confidence, and meaningful career opportunities.

Our **Programs**



Business Analytics

Turn data into strategic insights and make better business decisions.

+ AGENTIC AI INTEGRATION



Data Analytics

Extract meaningful insights, solve real-world problems, and drive business impact.

+ AGENTIC AI INTEGRATION



Data Science

Build intelligent solutions and apply data-driven thinking to solve real-world challenges.

+ ADVANCE AI INTEGRATION



SAP

Master enterprise solutions and strengthen your expertise in global business operations.

+ AI INTEGRATION



Cybersecurity

Protect systems, manage digital risks, and build a stronger foundation in cybersecurity.

+ AI INTEGRATION



Investment Banking

Develop financial expertise, analyze opportunities, and create value in the world of investment banking.

+ AI INTEGRATION

Ready to **Start?**

Check your eligibility

APPLY NOW

Connect with your admission counselor here:



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